Olist End-to-End Business Analytics Decision Platform

DEPI Graduation Project. Phase 1-3 Documentation Pack: Project Planning and Management, Literature Review, Requirements Gathering. Dataset figures were recomputed from the nine source CSV files on 29 Oct 2026. Anything marked 'proposed' is a team target or plan, not a measured result.

Repository: https://github.com/HadyAbohany/End-to-End-Business-Analytics-Decision-Platform

Part 1. Project Planning and Management

1. Project Proposal

Project title: End-to-End Business Analytics Decision Platform (Olist Brazilian E-Commerce data)

1.1 Project Overview

Olist is a Brazilian marketplace that connects independent sellers to customers. Its operational data is spread over nine tables (orders, items, payments, reviews, customers, products, sellers, category translation, geolocation) and no integrated solution exists to monitor performance, explain its drivers and predict outcomes. The dataset covers 99,441 orders from 4 Sep 2016 to 17 Oct 2018 and shows three business problems:

- **Retention:** 93,099 of 96,096 unique customers (96.9%) bought only once.
- **Delivery reliability:** 8.11% of delivered orders arrive after the promised date, and satisfaction collapses when they do (4.29 stars on time, 2.57 when late, 1.73 when more than 7 days late).
- **Delivery promises:** Olist promises 23.7 days on average while orders actually arrive in 12.6 days, so its promise misses the real delivery time by 12.7 days on average (mean absolute error).

This project follows the CRISP-DM lifecycle to build a complete solution: a relational database, a documented cleaning pipeline, exploratory analysis, an interactive dashboard, three machine-learning models, and a deployed prediction service.

1.2 Objectives

General objective: build an end-to-end data science solution that integrates the Olist data sources, monitors business performance, explains the drivers of late delivery, low satisfaction and low retention, and predicts future outcomes for decision makers.

Specific objectives

1. Load all nine tables into a relational database (Bronze, Silver, Gold layers) and enrich it with Brazilian public holidays from a REST API.
2. Implement and test a 19-rule data-quality register so that every join preserves its grain and item sales (BRL 13,591,643.70), freight (BRL 2,251,909.54) and payments (BRL 16,008,872.12) reconcile exactly.
3. Run exploratory analysis of sales, geography, delivery performance, seller SLA, reviews and retention.
4. Build an interactive dashboard with executive KPIs, trends, comparisons and a geographic view.
5. Train and compare models on a time-ordered split: Model A (late-delivery classification), Model B (delivery-time regression and quantile promise), Model C (customer segmentation).
6. Serve the models through a FastAPI service tracked with MLflow and packaged in Docker.
7. Communicate results through an executive summary, technical report, dashboard and stakeholder presentation.

1.3 Scope

In scope

- SQL database, data collection from CSV and one REST API (public holidays), cleaning, integration, normalisation and categorical transformation.
- EDA: descriptive statistics, distributions, correlations and group-by analysis.
- Dashboard pages: executive KPIs, logistics and SLA, product and category, customer retention, scenario simulator.
- Models A, B and C with baselines, model comparison, limitations and selection rationale.
- FastAPI, MLflow and Docker (the project's advanced track).

Out of scope

- Profit and margin analysis (the data has no cost of goods or carrier tariffs).
- Live integration with Olist systems or carrier tracking.

- Running real interventions (vouchers, new delivery promises); these are evaluated as scenarios only.
- GitHub Actions and cloud deployment unless the core is finished by the 23 Nov checkpoint (optional extension).

1.4 Deliverables

A repository containing: Business Case, Data Sources, SQL scripts, data collection (REST API script), data cleaning pipeline, EDA notebook, visualisations, machine-learning code, model evaluation report, dashboard, deployment files (FastAPI, MLflow, Docker), documentation (README, data dictionary, test plan, bug reports, user manual) and the stakeholder presentation, plus the DEPI documentation and infographic.

1.5 Assumptions and Constraints

Assumptions: the public Olist data is representative of a multi-seller marketplace; zip-prefix centroids give usable distance estimates; only information available at the stated prediction time is used for each model.

Constraints: fixed DEPI deadlines (design 6 Nov, implementation 30 Nov, final submission 4 Dec 2026); team of six; free tools only; data covers 2016-2018, with usable monthly volume only from Jan 2017 to Aug 2018.

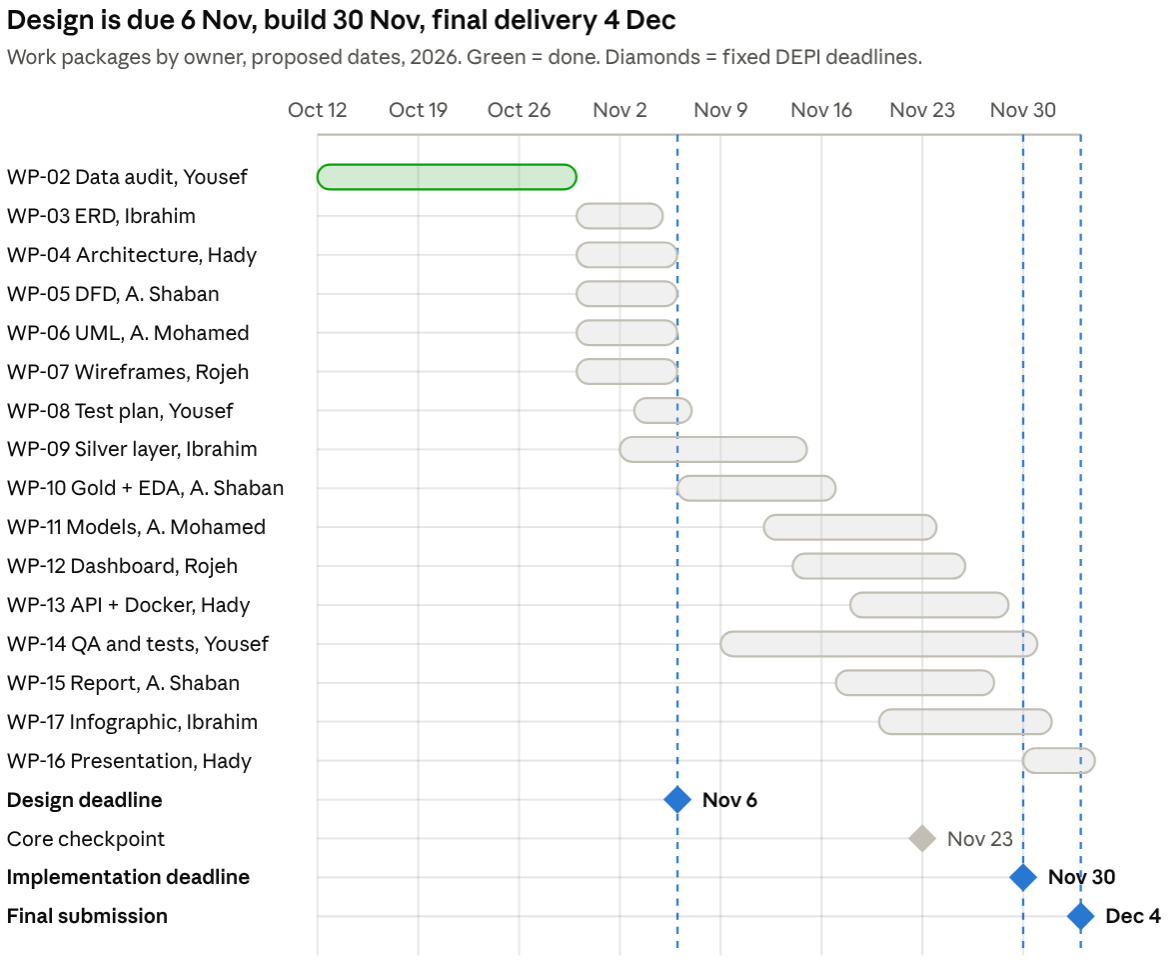
2. Project Plan

The plan is built backwards from the three fixed DEPI deadlines. Dates after 30 Oct are proposed and should be confirmed by the team.

2.1 Phases and Milestones

Phase	Window	Main work	Milestone
A. Planning and analysis	12 Oct - 30 Oct	Phase 1-3 documents; audit of the nine tables and the 19-rule quality register (completed 29 Oct)	Phase 1-3 pack
B. System analysis and design	30 Oct - 6 Nov	ERD and schema, DFD, architecture, sequence/activity/state/class diagrams, wireframes, technology stack	6 Nov: design deadline (fixed)

Phase	Window	Main work	Milestone
C. Implementation	2 Nov - 30 Nov	Bronze load and holiday API, Silver and Gold layers, EDA, three models, dashboard, FastAPI and MLflow, testing and documentation	23 Nov: core checkpoint; 30 Nov: implementation deadline (fixed)
D. Final delivery	30 Nov - 4 Dec	Final presentation, live demo, infographic, repository clean-up	4 Dec: final submission (fixed)



Gantt chart · 16 work packages, 4 milestones

Bars are work packages by owner; the dashed lines are the three fixed DEPI deadlines. All dates after 30 Oct are proposed.

The Silver layer (T6) gates modelling and the dashboard, so it carries the highest schedule risk. At the 23 Nov checkpoint the team decides whether the optional extension (GitHub Actions and cloud deployment) goes ahead.

2.2 Resources

Resource	Plan
People	Six team members, one owner per area (section 3)
Data	Olist Brazilian E-Commerce Public Dataset (nine CSV files) and the Nager.Date public holidays REST API
Database	PostgreSQL (free tier or local), with SQLite as a local fallback
Analysis and modelling	Python, pandas, scikit-learn, a gradient-boosting library, MLflow
Dashboard	Streamlit or Power BI, chosen by team skills
Serving	FastAPI and Docker
Collaboration	One GitHub repository, feature branches, review swap, one commit history per member
Budget	None: free tools only

3. Task Assignment and Roles

Each member owns two Phase 1-3 documents, one area of the build and one review. Roles are proposed and should be confirmed by the team.

3.1 Team Roles

Member	Proposed role	Phase 1-3 documents	Build responsibilities	Reviews work of
Hady Ahmed Abdelhady Mahmoud (team leader)	Project lead, FastAPI and frontend	Project Proposal; Project Plan	Repository, schedule, FastAPI service, MLflow, Docker, frontend, final presentation	Yousef Sayed
Yousef Sayed Ashour Mohamed	QA and data audit	Risk Assessment; KPIs	Data-quality register, reconciliation tests, test plan, bug reports, leakage review, user manual	Rogy Tamer

Member	Proposed role	Phase 1-3 documents	Build responsibilities	Reviews work of
Rogy Tamer Raouf Hanna	BI dashboard and visualisation	Functional Requirements; Non-functional Requirements	Wireframes, dashboard, scenario simulator, infographic	Ibrahim Mohamed
Ibrahim Mohamed Ibrahim Desouky	Data engineer and database	Task Assignment and Roles; Final Grading Criteria	ERD and schema, PostgreSQL load, holiday API script, Silver layer	Ahmed Shaban
Ahmed Shaban Sadiq Gomaa	Data analyst and feature engineering	Stakeholder Analysis; Suggested Improvements	Gold tables, EDA, distance features, business recommendations	Ahmed Mohamed
Ahmed Mohamed Abdelsalam Mohamed	Machine-learning engineer	User Stories and Use Cases; Feedback and Evaluation	Time-ordered splits, Models A, B, C, evaluation and model cards	Hady Ahmed

3.2 Task Breakdown

ID	Task	Owner	Support	Due	Output
T1	Phase 1-3 documents	Each member (2 each)	Review swap	Phase 1-3 deadline	12 documents in the repository
T2	Data audit and 19-rule quality register	Yousef	Ahmed Shaban	29 Oct (done)	Data Exploration Report
T3	Schema design and ERD	Ibrahim	Ahmed Shaban	3 Nov	ERD, DDL scripts
T4	Architecture, DFD, sequence/activity/state/class diagrams, wireframes	Hady and Rogy	Yousef	6 Nov	Design pack
T5	Bronze load and holiday API	Ibrahim	Yousef	9 Nov	Loaded database, ingestion script
T6	Silver and Gold layers, features	Ibrahim and Ahmed Shaban	Yousef	14 Nov	Three Gold tables, reconciliation tests
T7	Models A, B, C	Ahmed Mohamed	Hady	23 Nov	Models, comparison tables, model cards

ID	Task	Owner	Support	Due	Output
T8	Dashboard and simulator	Rogy	Ahmed Shaban	25 Nov	Running dashboard
T9	FastAPI, MLflow, Docker	Hady	Ahmed Mohamed	28 Nov	Running service
T10	QA, tests, bug reports, user manual	Yousef	All	30 Nov	Test pack
T11	Presentation, demo, infographic	Hady and Rogy	All	4 Dec	Slides, demo, infographic

3.3 Working Rules

- One template (font, title block, format) for all documents, numbered files in one folder per phase.
- Review swap before every submission; each member commits their own files.
- Feature branches merged by pull request, with the reviewer named in the description.
- Personal contact details (emails, phone numbers) are not stored in the public repository.

4. Risk Assessment and Mitigation Plan

Twelve risks are tracked. Figures come from the verified data audit.

ID	Risk	Category	Severity	Mitigation
R1	Join fan-out: joining orders to items (112,650 rows), payments (103,886), reviews (99,224) or raw geolocation (1,000,163) inflates revenue and duplicates orders	Data	Critical	Aggregate each table to one row per order_id before joining; reduce geolocation to 19,015 zip-prefix centroids; assert grain and reconcile money after every join
R2	Temporal instability: monthly late rate swings from 1.4% (Jun 2018) to 21.4% (Mar 2018), so random splits give misleading scores	Modelling	High	Time-ordered train, validation and test periods; evaluate on rolling folds; recalibrate thresholds

ID	Risk	Category	Severity	Mitigation
R3	Class imbalance: only 8.11% of delivered orders are late, so an always-on-time model scores 91.9% accuracy	Modelling	High	Use PR-AUC, recall at chosen precision and F1; class weights; no accuracy-only reporting
R4	Leakage: delivery dates, carrier date and review score are known only after checkout	Modelling	Critical	Declare the prediction time for each model; build model-specific feature views; compute seller history only from earlier orders
R5	Data quality: 1,359 orders with carrier date before approval, 23 delivered before carrier handover, 775 orders without items, 4 products at 0 g	Data	Medium	Flag, never silently clip; document each treatment in the 19-rule register
R6	Identity confusion: customer_id (99,441 values) is a per-order token; customer_unique_id (96,096) is the person	Analytics	High	Use customer_unique_id for every retention and RFM metric
R7	Frequency collapse: 96.9% of customers have one order, so K-Means on raw frequency gives degenerate clusters	Modelling	Medium	Use recency, log monetary value, review score and a repeat-buyer flag; compare with a rule-based RFM grid
R8	Multi-seller orders (1,278) make seller SLA and complaint attribution ambiguous	Data	Medium	Keep an item-level table; exclude multi-seller orders from seller-level complaint analysis
R9	Incomplete edge months (2016: 329 orders; Sep-Oct 2018: 20) distort trends	BI	Medium	Trend charts use Jan 2017 to Aug 2018 and state the reason
R10	Dashboard and API give different answers	Integration	Medium	Both read the same Gold views and the same registered model version

ID	Risk	Category	Severity	Mitigation
R11	Privacy: exposure of identifiers or personal contact details	Security	Low	Use anonymised customer_unique_id, aggregate maps at state level, keep secrets in environment variables, no personal data in the repository
R12	Scope and schedule: three models, dashboard, API and Docker within 4 weeks of implementation	Project	High	Core-first order, 23 Nov checkpoint, drop the optional extension if the core is late, one owner per area

5. Key Performance Indicators (KPIs)

Targets marked proposed will be revised after the baseline runs. Verified baselines come from the data audit.

5.1 Data KPIs

KPI	Target	Measured by
Row counts loaded vs source CSVs	100% match for nine tables	Automated row-count test
Money reconciliation (item sales, freight, payments)	0.00 BRL unexplained difference to the figures above	Reconciliation test after each join
Quality rules implemented and tested	19 of 19	Silver pipeline test log
Duplicate keys at declared grain in Gold tables	0	Uniqueness assertions

5.2 Model KPIs

KPI	Verified baseline	Proposed target	Evaluation
Model A: PR-AUC (late delivery)	0.081 (positive rate)	At least 2x the positive rate	Out-of-time test period

KPI	Verified baseline	Proposed target	Evaluation
Model A: ROC-AUC	0.50 (random)	At least 0.70	Out-of-time test period
Model A: recall at chosen precision	not applicable	Reported with threshold and cost reasoning	Validation period
Model B: MAE (days)	Olist promise 10.72 on the first-pass test window; median baseline 5.08	At most 4.0 (first pass: 3.59)	Out-of-time test period
Model B: quantile promise coverage	Olist 93.4% at 18.5 days	Measured coverage within 2 points of nominal; shorter promise than Olist at equal coverage	Rolling time folds
Model C: silhouette and stability	not applicable	Silhouette at least 0.25 with interpretable segments	Customer snapshot

5.3 System and Project KPIs

KPI	Target	Measured by
Dashboard pages	5 of 5 working	Dashboard checklist
Dashboard load time	5 seconds or less locally	Browser test
API response time (single prediction)	Under 500 ms (proposed)	API test
Automated tests passing	100%	Test run
Members with commits	6 of 6	Commit history
Documents peer-reviewed	12 of 12	Review notes

Part 2. Literature Review

1. Introduction

A marketplace such as Olist depends on three things at once: reliable logistics, customer satisfaction and repeat purchases. Reichheld and Sasser (1990) argued that keeping existing customers is a major profit driver for service businesses. In the Olist data this

matters directly: 96.9% of customers bought once, and late delivery cuts the average review score from 4.29 to 2.57. This review covers prior work behind our technical choices: delivery-time estimation with quantile models, evaluation under class imbalance and time drift, customer segmentation, explainability, and data-preparation practice. It then reviews the dataset and states how the project builds on the literature.

Note on sources: the references below are established methods papers and books. We did not find a peer-reviewed benchmark on the Olist data that we can cite, so our comparison baseline is Olist's own delivery promise, measured on the dataset. All citations should be checked against the originals before final submission.

2. Related Work

2.1 Delivery-time prediction and promise calibration

A delivery promise is a statement about the upper part of the delivery-time distribution, not its mean. Koenker and Bassett (1978) introduced quantile regression, which estimates a chosen conditional quantile directly, and Meinshausen (2006) extended the idea to random forests. Romano, Patterson and Candes (2019) proposed conformalized quantile regression, which adds a calibration step to give coverage guarantees under an exchangeability assumption. Delivery data that drifts over time breaks that assumption, so our project measures coverage on later periods instead of relying on the guarantee. For the base learners we use gradient-boosted trees (Chen and Guestrin, 2016; Ke et al., 2017), which are widely used for tabular data.

In our first-pass experiment (time-ordered split, 14,399 test orders from 21 Jun to 29 Aug 2018) a boosted model reached a mean absolute error of 3.59 days, against 10.72 days for Olist's promised date and 5.08 days for a median baseline. A quantile model at 0.90 delivered 95.1% coverage with a 16.8-day average promise, against 93.4% coverage and 18.5 days for Olist. These are first-pass results from a two-month window and will be re-checked on rolling folds.

2.2 Class imbalance and evaluation metrics

Only 8.11% of delivered orders are late. Chawla et al. (2002) proposed SMOTE for synthetic minority oversampling, and He and Garcia (2009) survey imbalanced learning. Saito and Rehmsmeier (2015) show that precision-recall curves are more informative than ROC curves when the positive class is rare. Our plan therefore uses class weights first, PR-AUC and recall at a chosen precision as main metrics, and resampling only inside the training period.

2.3 Time-ordered validation

Tashman (2000) reviews out-of-sample evaluation for forecasting, and Bergmeir and Benitez (2012) compare cross-validation schemes for time-series predictors. In the Olist data the monthly late rate moves between 1.4% (Jun 2018) and 21.4% (Mar 2018), so we evaluate on later periods and report results on rolling time folds instead of a random split.

2.4 Customer segmentation

RFM analysis (recency, frequency, monetary value) goes back to Hughes (1994). It assumes that customers buy repeatedly, but 93,099 of 96,096 Olist customers have exactly one order, so frequency barely varies. We keep recency and log monetary value, add mean review score, and use frequency as a repeat-buyer flag. Cluster quality is judged with the silhouette coefficient (Rousseeuw, 1987), stability across seeds and business meaning.

2.5 Explainability and process

Lundberg and Lee (2017) introduced SHAP values, which attribute a prediction to its input features. We use them to show why an order is flagged as high-risk. The overall lifecycle follows CRISP-DM (Wirth and Hipp, 2000), and the table design follows the dimensional-modelling principle of stating the grain of each table before joining (Kimball and Ross).

2.6 Summary of the Literature

Source	Topic	What it contributes	Use in this project
Reichheld and Sasser (1990)	Retention economics	Links retention to profit in service businesses	Business case for retention analysis
Koenker and Bassett (1978)	Quantile regression	Estimates conditional quantiles directly	Delivery promise at a chosen coverage
Meinshausen (2006)	Quantile regression forests	Conditional distribution from random forests	Alternative quantile model
Romano, Patterson and Candes (2019)	Conformal quantile regression	Calibrated prediction intervals under exchangeability	Calibration idea, checked on later periods
Chen and Guestrin (2016); Ke et al. (2017)	Gradient boosting	Fast, strong tabular learners	Models A and B

Source	Topic	What it contributes	Use in this project
Chawla et al. (2002); He and Garcia (2009)	Imbalanced learning	Resampling and survey of methods	Late class is 8.11%
Saito and Rehmsmeier (2015)	Evaluation	PR curves are more informative on rare classes	PR-AUC as main metric
Tashman (2000); Bergmeir and Benitez (2012)	Validation	Out-of-sample and time-series evaluation	Time-ordered splits
Hughes (1994); Rousseeuw (1987)	Segmentation	RFM concept; silhouette score	Model C
Lundberg and Lee (2017)	Explainability	SHAP feature attributions	Reasons shown with each risk score
Wirth and Hipp (2000)	Process	CRISP-DM lifecycle	Project phases

3. Dataset Review

The project uses the Olist Brazilian E-Commerce Public Dataset (Olist and Sionek, 2018): nine CSV files, 99,441 orders from 4 Sep 2016 to 17 Oct 2018, plus public holidays from the Nager.Date REST API. All counts below were recomputed from the files.

Table	Rows	Verified notes
orders	99,441	96,478 delivered; 160 missing approval, 1,783 missing carrier and 2,965 missing delivery timestamps; 1,359 carrier dates before approval
order_items	112,650	98,666 orders, 32,951 products, 3,095 sellers; 775 orders have no items; 1,278 orders have several sellers
order_payments	103,886	99,440 orders; credit card 78.3% of value, boleto 17.9%; 9 zero-value rows
order_reviews	99,224	98,673 orders; 547 orders have several reviews; 768 orders have none; 40,977 written messages
customers	99,441	96,096 unique customers; 27 states

Table	Rows	Verified notes
products	32,951	610 missing categories; 4 products with 0 g weight
sellers	3,095	59.74% in SP; top 5% of sellers earn 53.2% of item sales
category_translation	71	2 categories missing: pc_gamer and portateis_cozinha_e_preparadores_de_alimentos
geolocation	1,000,163	19,015 zip prefixes; 261,831 exact duplicates; 42 rows outside Brazil

Limits to state openly: no cost or margin data, almost no orders in 2016 and after Aug 2018, reviews exist only for customers who respond, and distances are centroid-based straight lines.

4. Research Gap and Project Positioning

Public work on e-commerce delivery and churn tends to stop at a notebook with a single score. This project adds the surrounding engineering: grain-safe SQL layers with reconciliation tests, labels and features tied to a declared prediction time, time-ordered evaluation, a promise policy compared with the company's own promise, and a deployed service and dashboard. It does not claim a new modelling method.

5. Suggested Improvements

These extend the core system. Each is a proposal that needs validation.

1. **Dynamic delivery promise.** Replace the single average buffer with route-specific quantile promises. First-pass evidence: a shorter promise at similar coverage, which must be confirmed on rolling folds. A promise of 14 days at 95% coverage is not supported by the data. Benefits: logistics and marketing teams.
2. **48-hour seller early warning.** At 48 hours after purchase, flag orders whose carrier handover is not recorded. Orders handed to the carrier after the shipping limit date are late 24.4% of the time, against 6.49% otherwise (association, not proof of cause). The voucher or apology action is a cost-benefit scenario, because the data holds no intervention results. Benefits: seller operations and customer support.
3. **Holiday enrichment through a REST API.** Fetch Brazilian public holidays for 2016, 2017 and 2018 and compute holidays during the expected transit window from purchase date and predicted duration, never from the actual delivery date. Keep the feature only if it improves out-of-time results. Benefits: data science team.

4. **Scenario simulator.** Let a manager change the distance or seller-location assumption and see model estimates for delivery days, freight and late risk. The data has no inventory or warehouse information, so results are model estimates, not causal effects. Benefits: logistics and finance.
5. **Seller complaint cards from review text.** Group the 40,977 Portuguese messages into delay, product problem and missing items, starting with keyword rules and a hand-labelled sample of 300 to 500 messages, and report precision and recall. Use only single-seller orders. Benefits: seller operations and quality.
6. **First-to-second purchase analysis.** Study which first-order conditions (delivery delay, review score, category, freight share) relate to a second purchase, since only 3.12% of customers return. Benefits: marketing and retention.

6. Feedback and Evaluation

6.1 Self-evaluation framework

Area	Check	Target	Evidence	Owner
Data integrity	Row counts and money totals match across layers	100% match, 19 of 19 rules tested	Reconciliation test log	Yousef, Ibrahim
Model validity	Metrics on later periods, no leaked columns	See KPI table (section 5.2)	MLflow runs, model cards	Ahmed Mohamed, Yousef
Dashboard	Five pages load and match Gold views	Load time 5 seconds or less	Dashboard checklist	Rogy
API and deployment	Endpoints answer correctly inside Docker	Under 500 ms, all tests pass	API test suite	Hady
Reproducibility	A fresh clone runs from the README	No manual fixes	Fresh-clone test	Yousef
Business usefulness	A stakeholder understands findings and actions	Feedback recorded and acted on	Walkthrough notes	Hady

6.2 Feedback plan

Source	Method	When	What we ask
Trainer	Review of the Phase 1-3 pack and then the design pack	After submission; before 6 Nov	Do scope, structure and grading mapping match expectations
Peers	Review swap (section 3.1)	Before each submission	Is the work consistent with the other documents
Simulated stakeholder	A member or the trainer plays the business user	Mid to late Nov	Are findings and recommendations clear and justified
Dashboard users	Three to five people outside the team do the dashboard tasks	After the dashboard is built	Can they find KPIs, delay drivers and the segment view unaided
Demo rehearsal	Internal run-through	1 to 3 Dec	Timing and clarity

6.3 Feedback log

No trainer feedback has been received yet. The team adds a row (date, source, feedback, action taken, status) each time feedback arrives.

6.4 Final evaluation report

After delivery the team writes a short report with results against each target, what worked and what did not, limitations (Section 3, last paragraph), and which suggested improvements to do first.

7. Final Grading Criteria

Marks as listed for DEPI projects; confirm with the trainer before submission.

Criterion	Marks	Where it is covered
1. Idea and objectives	10	Part 1, section 1
2. System analysis, requirements and design	15	Part 3; design pack due 6 Nov
3. Algorithms, methods and techniques	20	SQL layers, cleaning, EDA, Models A, B, C (tasks T3 to T7)

Criterion	Marks	Where it is covered
4. Implementation, coding and results	20	Pipeline, dashboard, API, MLflow (tasks T5 to T9)
5. Testing and documentation	10	Task T10: test plan, tests, bug reports, README, user manual
6. Innovation and added value	10	Suggested improvements 1 to 6 above
7. Presentation and demo	10	Task T11: slides, live demo, infographic
8. Final submission and delivery	5	Complete repository, on-time uploads, dashboard link

Open point: criterion 8 refers to a deployed dashboard link. The team should choose a free hosting option or confirm with the trainer what counts as deployed.

8. References

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Part 3. Requirements Gathering

3.1 Stakeholder Analysis

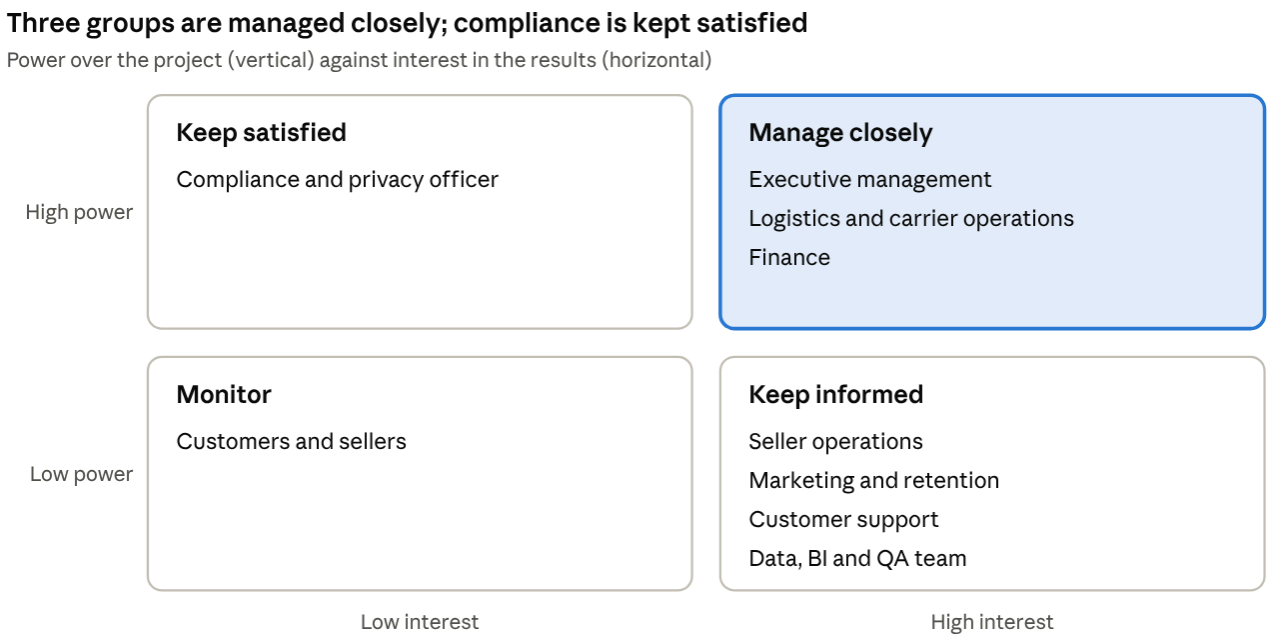
Nine stakeholder groups are affected by the platform. Executives, logistics and finance hold the most power; seller operations, marketing and support are the main users.

3.1.1 Stakeholder register

Stakeholder	Role	Needs from the system	Concerns	Power	Interest
Executive management	Sponsor, strategy	Verified KPIs (orders, item sales, payments, review score, repeat rate) and evidence of ROI	Inflated numbers from bad joins; black-box models	High	High
Logistics and carrier operations	Delivery promises and routes	Delay risk, delivery-time estimates, state and route bottlenecks	Promises that cause late deliveries; high cross-state freight	High	High
Finance	Payments and cost control	Separate item sales, freight and payments; reconciled totals; cost of any intervention	Spending on vouchers without proven benefit; unexplained payment gaps	High	Medium
Seller operations	Monitors 3,095 sellers	Seller SLA breach rates, early-warning list, complaint themes	Unfair penalties on sparse data or multi-seller orders	Medium	High
Marketing and retention	Customer growth	Customer segments, first-to-second purchase insight	Targeting unhappy or low-value segments	Medium	High

Stakeholder	Role	Needs from the system	Concerns	Power	Interest
Customer support	Handles complaints	Early list of orders likely to arrive late	Ticket spikes after long delays	Low	High
Data, BI and QA team	Builds and maintains the platform	Clean layers, reproducible pipelines, tests	Leakage, drift, schema changes	Medium	High
Compliance and privacy officer	Lawful data use	Anonymised identifiers, aggregated locations, access control	Exposure of personal data	High	Low
Customers and sellers	External users	Reliable delivery dates; fair seller evaluation	Misleading promises; opaque scoring	Low	Low

3.1.2 Power/interest grid



power/interest grid · 9 stakeholders in 4 zones

3.1.3 Engagement strategy

- **Manage closely:** short findings summary with verified KPIs; agree what counts as late, high-risk and satisfied before modelling.
- **Keep satisfied:** consult early on which fields are shown; confirm that no personal identifiers appear in dashboards or APIs.
- **Keep informed:** dashboard walkthrough, early-warning list review with seller operations and support.

- **Monitor:** design outputs (promises, seller scores) so they are accurate and fair to external users.

3.1.4 Conflicts and resolutions

1. **Short promise vs safe promise.** Marketing wants a shorter delivery promise at checkout; logistics wants a buffer to avoid late orders. Resolution: route-specific quantile promises (Model B), judged on measured coverage on later periods; first-pass evidence shows about 2 to 3 days shorter at similar coverage, to be confirmed.
2. **Proactive vouchers vs cost control.** Support wants to compensate likely-late orders; finance wants to limit spend. Resolution: treat vouchers as a cost-benefit scenario with an alert threshold chosen from the cost of a missed delay against a false alarm.
3. **Accuracy vs interpretability.** Data scientists prefer boosted trees; executives need explanations. Resolution: compare with simple baselines and show SHAP reasons next to each risk score.
4. **Seller accountability vs fair evidence.** Seller operations want to act on SLA breaches, but the link between breach and lateness is an association and many orders have several sellers. Resolution: show sample sizes, exclude multi-seller orders from seller-level scoring, and require review before any penalty.

3.1.5 Stakeholder needs mapped to features

Need	Feature
Big picture and trust in numbers	Executive KPI page with reconciled totals
Delay risk and promise accuracy	Models A and B, API endpoints, logistics page
Seller follow-up	48-hour early-warning list, seller leaderboard
Customer targeting	Model C segments, first-to-second purchase analysis
Cost control	Scenario simulator and cost-benefit view
Privacy	Anonymised identifiers, state-level maps

3.2 User Stories and Use Cases

3.2.1 Actors

Actor	Description
Data engineer	Loads data, builds Bronze, Silver and Gold layers, runs the REST API ingestion
Data scientist	Builds features, trains and evaluates Models A, B and C
QA engineer	Runs reconciliation and leakage checks, writes tests and bug reports
Executive / finance lead	Reads KPIs and reconciled totals
Logistics and seller-operations manager	Uses delay risk, early warning and the scenario simulator
Marketing and retention specialist	Uses segments and retention analysis
API client	An application that calls the prediction endpoints

3.2.2 User stories (MoSCoW)

ID	User story	Acceptance criteria	Requirements	Priority
US-01	As a data engineer, I want all nine tables and holiday data in a relational database so that analysis uses one source of truth	Row counts match the CSVs; no orphan keys; holiday table filled for 2016 to 2018	FR-01, FR-02	Must
US-02	As a QA engineer, I want a tested 19-rule cleaning register so that bad records never distort metrics	Each rule has a count, a treatment and a test; geolocation reduced to 19,015 centroids	FR-03, NFR-04	Must
US-03	As a data analyst, I want Gold tables at order, item and customer grain so that joins never inflate revenue	Order table has 99,441 rows; item sales sum to BRL 13,591,643.70; 96,096 unique customers	FR-04	Must

ID	User story	Acceptance criteria	Requirements	Priority
US-04	As an analyst, I want exploratory analysis of sales, delivery, sellers and reviews so that drivers are understood	Notebook with descriptive statistics, distributions, correlations and group-by results	FR-05	Must
US-05	As a logistics manager, I want a late-delivery risk model so that risky orders are flagged early	Time-ordered evaluation; PR-AUC, recall and calibration reported; no post-checkout columns	FR-06, NFR-05	Must
US-06	As a product manager, I want a delivery-time and promise model so that promises are accurate	MAE compared with Olist's promise; coverage measured on later periods	FR-07	Must
US-07	As a seller-operations manager, I want a 48-hour early-warning list so that I can contact sellers before orders are late	Uses only information known at 48 hours; lists sample sizes	FR-08	Should
US-08	As a marketing specialist, I want customer segments so that campaigns target the right groups	Segments from recency, log monetary value, review score and repeat flag; silhouette and stability reported	FR-09	Must
US-09	As an executive, I want an overview page with reconciled KPIs so that I trust the numbers	Item sales, freight and payments shown separately; definitions in tooltips	FR-10, NFR-01	Must
US-10	As a logistics manager, I want delivery and SLA views by state and distance so that I can find bottlenecks	Same-state vs cross-state comparison; late rate by state	FR-11	Must
US-11	As a manager, I want a scenario simulator so that I can test distance or location assumptions	Outputs labelled as model estimates with stated assumptions	FR-12	Should

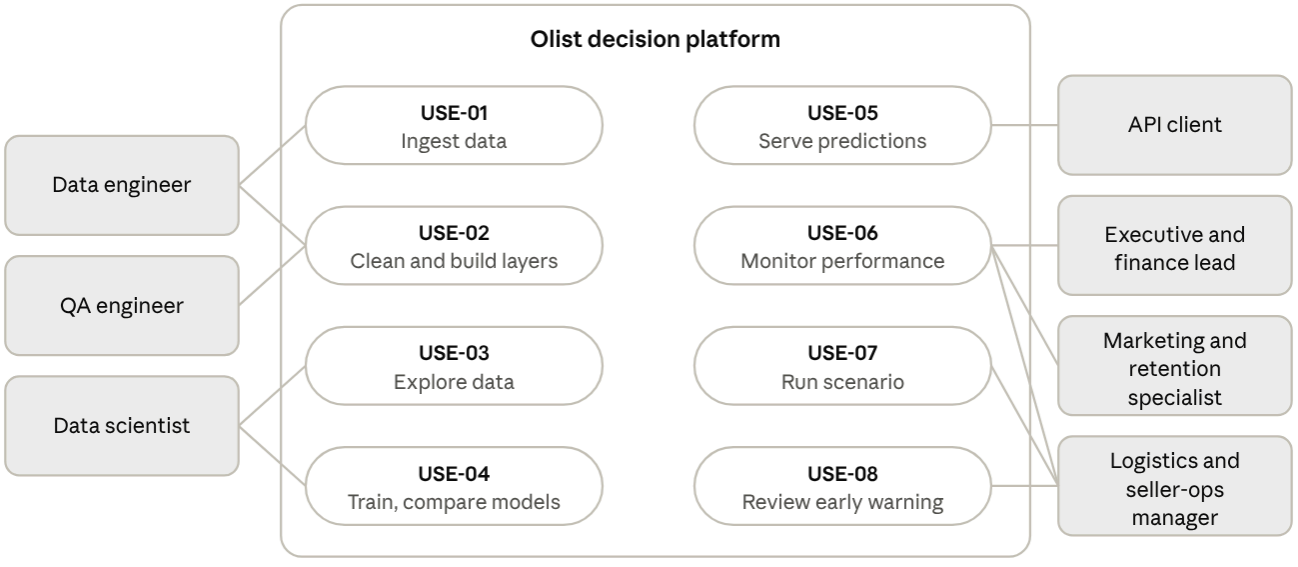
ID	User story	Acceptance criteria	Requirements	Priority
US-12	As an API client, I want prediction endpoints so that applications can use the models	Documented endpoints answer in under 500 ms inside Docker	FR-13, NFR-02	Must
US-13	As a quality auditor, I want seller complaint themes from review text so that problems are explained	Precision and recall reported on a hand-labelled sample	FR-14	Could
US-14	As a compliance officer, I want no personal identifiers exposed so that data use is lawful	Dashboards and APIs show only anonymised ids and aggregates	NFR-06	Must
US-15	As a new team member, I want setup instructions so that I can reproduce results	A fresh clone runs end to end from the README	FR-15, NFR-10	Should

3.2.3 Use cases

ID	Use case	Actor	Main flow	Alternative flow	Requirements
UC-01	Ingest data	Data engineer	Create tables; load nine CSVs; call the holiday API for 2016 to 2018; check row counts	If a count differs or the API fails, abort, log, and use the cached holiday snapshot	FR-01, FR-02
UC-02	Clean and build layers	Data engineer, QA engineer	Apply the 19 rules; build centroids; aggregate items, payments, reviews to order grain; run reconciliation tests	If money does not reconcile, stop and report the join that caused it	FR-03, FR-04
UC-03	Explore data	Data scientist	Compute statistics, distributions, correlations and group-by results; record findings	Months with too little data are excluded from trends	FR-05

ID	Use case	Actor	Main flow	Alternative flow	Requirements
UC-04	Train and compare models	Data scientist	Split by time; build model-specific feature views; train baselines and candidates; log runs in MLflow	If a feature is unavailable at prediction time, remove it and re-run	FR-06, FR-07, FR-09, NFR-05
UC-05	Serve predictions	API client	Send order features; receive late probability, estimated days and promise with reasons	Invalid input returns a clear error	FR-13, NFR-02
UC-06	Monitor performance	Executive, logistics manager	Open the dashboard; filter dates and statuses; read KPIs and maps	If a month is incomplete, a note explains the exclusion	FR-10, FR-11
UC-07	Run scenario	Logistics manager	Choose assumptions; the simulator returns estimates with limits	Out-of-range assumptions are rejected	FR-12
UC-08	Review early-warning list	Seller-operations manager	Open the 48-hour list; sort by risk; contact sellers	If the list is too long, raise the threshold	FR-08

Eight use cases connect seven actors to the platform



use-case diagram · 7 actors, 8 use cases

The use-case diagram, DFD and other diagrams are part of the design pack due on 6 Nov.

3.3 Functional Requirements

ID	Requirement	Specification
FR-01	Relational database	The system shall load all nine Olist tables into a relational database with declared keys and a Bronze, Silver and Gold structure
FR-02	REST API collection	The system shall fetch Brazilian public holidays for 2016 to 2018 from the Nager.Date API, store them with source and retrieval time, and fall back to a cached snapshot if the API is unavailable
FR-03	Data cleaning	The system shall apply the 19-rule register: flag timestamp anomalies, handle missing and zero values, translate categories, and reduce geolocation to one centroid per zip prefix inside Brazil's bounds
FR-04	Grain-safe Gold tables	The system shall build order, item and customer tables with unique declared keys, and reconcile item sales, freight and payments after every join
FR-05	Exploratory analysis	The system shall provide descriptive statistics, distributions, correlations and group-by analysis of sales, delivery, sellers, reviews and retention
FR-06	Late-delivery classification	The system shall train and compare classifiers for is_late on a time-ordered split and report PR-AUC, recall, F1, ROC-AUC and calibration against baselines
FR-07	Delivery-time and promise model	The system shall predict delivery days and quantile-based promises, and report MAE, RMSE and coverage against Olist's promise
FR-08	48-hour early warning	The system shall score orders without a recorded carrier handover at 48 hours, using only information available then
FR-09	Customer segmentation	The system shall segment customers by customer_unique_id with a fixed snapshot date and report silhouette score and stability

ID	Requirement	Specification
FR-10	Executive KPI page	The dashboard shall show orders, item sales, freight, payments, average order value, late rate, repeat rate and review score, each with a documented definition
FR-11	Logistics and category views	The dashboard shall show delivery performance and SLA by state, distance and seller, and category sales concentration
FR-12	Scenario simulator	The dashboard shall let users change distance or location assumptions and show model estimates with stated limits
FR-13	Prediction API	The system shall expose health, delay-risk, delivery-time and early-warning endpoints, with validated inputs and the model version in each response
FR-14	Complaint analysis	The system shall classify review messages from single-seller orders into complaint themes and report accuracy on a labelled sample
FR-15	Documentation and tests	The repository shall contain SQL scripts, data dictionary, notebooks, pipeline code, tests, bug reports, README and user manual

3.4 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Dashboard pages shall load within 5 seconds in local testing
NFR-02	Performance	The API shall answer a single prediction in under 500 ms (proposed target)
NFR-03	Usability	Charts shall have plain titles, consistent colours, visible filters and KPI definitions
NFR-04	Reliability and auditability	The pipeline shall never silently delete or clip records; every anomaly is flagged and counted
NFR-05	Scientific rigour	Imputers, scalers, encoders and seller-history features shall be fitted only on earlier data than the period they are applied to

ID	Category	Requirement
NFR-06	Security and privacy	Outputs shall use anonymised identifiers and aggregated locations; secrets are kept in environment variables
NFR-07	Maintainability	Code shall be modular (sql, etl, models, api, dashboard, tests), commented and style-checked
NFR-08	Portability	The service shall run in Docker with pinned dependencies
NFR-09	Scalability	Heavy aggregations shall run in the database, not in the dashboard
NFR-10	Reproducibility	Seeds, split dates, package versions and model versions shall be recorded so metrics can be reproduced
NFR-11	Compatibility	The dashboard shall work in current Chrome, Edge and Firefox
NFR-12	Interpretability	Predictions shall show their main drivers and state their limits